

Lam Fung Academy

SSPA Mock Exam (2) · Primary 5 Mathematics

Scope: Lessons 1–18 (Full Semester Comprehensive) — Multiple digits • Estimation • Area •
Fractions • Decimals • Algebra • Equations

Exam time: 75 minutes Full score: 100 points

Name: _____

date: _____

Fraction: _____ / 100

Part A: Multiple choice questions

Part A - Multiple Choice Questions

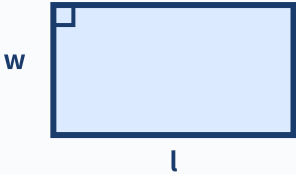
12 questions × 2.5 points = 30 points

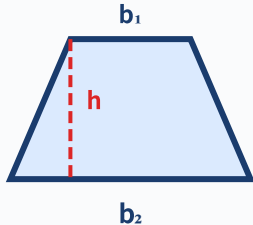
Instructions: Choose the correct answer and write the letter of your answer in the brackets to the right of each question. There is only one correct answer for each question. Note: Each question is worth 2.5 points, that is, every 2 questions is worth 5 points in total.

Exam tips (SSPA must read)

- ① Do the questions you know first, don't get stuck on one!
- ② Check the unit for each question (cm vs cm² vs cm³)!
- ③ Geometry questions: If there is an elevation in the picture, use the height. If there is no elevation, use the formula to find it!
- ④ Application question: Write an answer sentence! Points will be deducted for not writing steps!
- ⑤ 5 minutes left: Check if the MC has been filled in and if the unit is correct!

question number	Question	fraction
1	How many "zeros" do you need to pronounce the number 60,080,090 in Chinese? A. 0 pieces B. 1 piece C. 2 pieces D. 3 pieces	___ / 2.5 points
2	Calculate $0.35 \times 2.4 = ?$ Where should the decimal point be placed? A. 0.084 B. 0.84 C. 8.4 D. 84	___ / 2.5 points
3	Simplify: $3a + 5 + 2a - 4 = ?$ A. $5a + 1$ B. $6a$ C. $6a - 1$ D. $a + 9$	___ / 2.5 points
4	Solve the equation: $3x + 7 = 22$, $x = ?$ A. 3 B. 5 C. 7 D. 15	___ / 2.5 points
5	Calculation: $\frac{3}{8} \times \frac{4}{9} = ?$ Which of the following steps would make the calculation easier?	___ / 2.5 points

	A. First multiply the numerator and denominator, and then reduce B. First reduce 3 and 9, 4 and 8, and then multiply SEP C. Multiply after dividing D. Convert to decimal and then calculate	
6	▲ Rectangle (area unit conversion reference)  Rectangle area = 6 m^2. Expressed in cm^2, what is the equivalent? A. 600 cm^2 B. $6,000 \text{ cm}^2$ C. $60,000 \text{ cm}^2$ D. $600,000 \text{ cm}^2$	____/ 2.5 points

question number	Question	fraction
7	If you round 7.648 to one decimal place (tenths of a second), what is the result? A. 7.6 B. 7.7 C. 7.64 D. 7.65	___ / 2.5 points
8	calculate: $\frac{2}{3} + \frac{1}{4} - \frac{1}{6} = ?$ A. $\frac{5}{12}$ B. $\frac{3}{4}$ C. $\frac{7}{12}$ D. $\frac{11}{12}$	___ / 2.5 points
9	"3 times a number plus 5 equals 20." Which of the following equations correctly represents this relationship? A. $x + 3 + 5 = 20$ B. $3x + 5 = 20$ C. $3(x + 5) = 20$ D. $x \div 3 + 5 = 20$	___ / 2.5 points
10	▲ Trapezoid: hypotenuse 10cm (no need to calculate area!)  Trapezoid: upper base 8 cm, lower base 12 cm, hypotenuse 10 cm, height 6 cm. Which of the following values does not require when calculating the area? A. Upper bottom 8 cm B. Lower bottom 12 cm C. Bevel 10 cm D. Height 6 cm D. Height 6 cm	___ / 2.5 points
11	There are 60 pieces in a pack of sugar. I ate $\frac{1}{5}$ and gave the remaining $\frac{1}{4}$ to my sister. How many pills does the sister get? A. 3 capsules B. 9 capsules C. 12 capsules D. 15 capsules	___ / 2.5 points
12	▲ Rectangle 12×5cm + triangle (base 12cm, height 4cm)  The picture below is composed of a rectangle (length 12cm, width 5cm) and a triangle (base 12cm, height 4cm). Total area of combined graphics = ? A. 60 cm^2 B. 72 cm^2 C. 84 cm^2 D. 96 cm^2	___ / 2.5 points

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S8	Square perimeter 36 cm. Area = ? (Note: You cannot directly substitute the perimeter!)	
S9	A rectangle 2.5 m × 80 cm. Area = ? cm ² (unify the units first)	

Algebra and equation (total 5 questions)

#	Question	Difficulty
S10	Simplify: $4a + 7 + 3a - 2 = ?$	
S11	Simplify: $5(2x - 3) - 3(x - 4) = ?$	
S12	Solve the equation: $2x + 9 = 3x - 4$	
S13	Four times a number minus 7 equals 29. Let this number be x , make an equation and find x .	
S14	Solve the equation: $3(2x - 1) = 5(x + 2)$	

fractionanddecimal (total 4 question)

#	Question	Difficulty
S15	$\frac{5}{6} - \frac{1}{4} + \frac{1}{3} = ?$	
S16	Calculate $0.36 \times 2.5 = ?$ (write down the steps)	
S17	There are 90 pieces in a pack of sugar. Eat $\frac{1}{3}$ of it, and then give the remaining $\frac{1}{5}$ to your brother. How many pieces does the brother get? $\frac{1}{3}$ Share it with your younger brother. How many pieces does the brother get?	

S18	Approximate 6.7849 to (a) tenth place (b) percentile (c) Which one is closer to the original number?	

Cross-Topic comprehensive (total 3 question)

#	Question	Difficulty
S19	The length of a rectangle is 3 times its width. If the perimeter is 48 cm, find its area.	
S20	A trapezoid has the same height as a parallelogram. Upper base of trapezoid + lower base = 16 cm, base of parallelogram = 16 cm. The area of a parallelogram is 120 cm^2 . Trapezoidal area = ?	
S21	The base of a triangle increases by 4 cm, leaving the height unchanged. The new area is 12 cm^2 more than the original area. What is the height of the original triangle?	

 Post-exam review record:

Part A score: ____/30 Part B score: ____/40 Part C score: ____/30 Total score: ____/100

The question types that need to be strengthened most in this exam: _____; The traps that need to be reviewed: _____

Part B - Short Questions/Calculation Questions8 questions $\times$ 5 points = 40 points

Instructions: Write the complete calculation steps and answers in the answer area below each question. Answers must be reduced (if applicable) and accompanied by units.

question number	Question	fraction
1	(a) Write in Arabic numerals: eight thousand six hundred and three thousand two hundred. (b) Round the above numbers to the nearest million.	___/ 5 points
2	Calculate the following decimal multiplications and write down the calculation process: (a) $0.6 \times 0.25 = ?$ (b) $1.2 \times 3.5 = ?$	___/ 5 points
3	Calculation: $2\frac{1}{2} - \frac{3}{4} + \frac{1}{3} = ?$ (Hint: Mixed fraction $\rightarrow$ Improper fraction, then calculate by common fraction)	___/ 5 points
4	Base of triangle = 15 cm, area = 45 cm^2 . Seek high.	___/ 5 points

question number	Question	fraction
5	Solve the equation: $4(x-3) = 2x+8$ Write out the complete steps of removing brackets → moving terms → merging → solving.	___/ 5 points
6	(a) Round 23.476 to one decimal place (tenths). (b) Approximate 23.476 to two decimal places (percentile). (c) Which approximation is closer to the original number? Why?	___/ 5 points
7	Calculation: $1\frac{3}{5} \times \frac{5}{8} = ?$ (hint: mixed fraction → improper fraction, cross reduction, then multiply)	___/ 5 points
8	Simplify the algebraic expression: $5(2a-3b) - 2(3a-b) + 7b$ Write: remove the brackets → combine similar terms → the simplest result.	___/ 5 points

Part C - Application Problems3 questions $\times$ 10 points = 30 points

Instructions: Each question must list all calculation steps (formulation, solution, verification) and write a complete answer. The answer must be in units or reduced to the simplest form.

Question 1 (10 points) – Equations + Graphing Applications

Cross-topic

____ / 10 points

The length of a rectangle is 5 cm more than its width.

Given perimeter = 50 cm.

- (a) Let the width be w cm, expressing length algebraically. (1 point)
- (b) According to the perimeter formula $(\text{length} + \text{width}) \times 2 = \text{perimeter}$, list the equation. (2 points)
- (c) Solve the equation and find the width w . (3 points)
- (d) Find the length and then the area of the rectangle. (3 points)
- (e) Check calculation: Substitute back into the original equation and confirm whether the perimeter is equal to 50 cm. (1 point)

Answer area - please write all steps, equation derivation and complete answer sentences:

Question 2 (10 points) - Mixed application of fractions + decimals

Cross-topic

____ / 10 points

My mother bought 3 boxes of juice, each with a capacity of 1.5 liters (L).

She poured all the juice into a large bottle.

On the first day, I drank all the juice $\frac{1}{3}$.

The next day, drink another 0.8 L from the remaining juice.

- (a) How many liters of total juice are there? (1 point)
- (b) How many liters did you drink on the first day? How many liters are left? (3 points)
- (c) How many liters of juice are left after the second day? (3 points)
- (d) What fraction of the original juice is left in the end? (Expressed as a decimal or simplest fraction) (3 points)

Answer area - please write all calculation steps and complete answer sentences (with unit L):

Question 3 (10 points) – Comprehensive cross-topic challenge**Cross-topic****____ / 10 points**

The floor of the school auditorium is a trapezoid with an upper base of 18 m, a lower base of 24 m, and a height of 12 m.

The school plans to re-tile the floors.

- (a) What is the area of the auditorium floor in m^2 ? (2 points)
- (b) Round the area to the nearest hundred (in m^2). (2 points)
- (c) Each square floor tile has a side length of 0.5 m. What is the area in m^2 of each floor tile? (2 points)
- (d) How many floor tiles are needed to cover the entire auditorium (using the exact area)? (2 points)
- (e) If floor tiles with sides 60 cm are used instead, how many tiles are needed? (Hint: Convert units first) (2 points)

Answer area - please write all calculation steps and complete answer sentences:

— Complete the entire volume —

Total score: _____ / 100 points

Part A ____/30 + Part B ____/40 + Part C ____/30 = Total score ____/100

Signature of marking teacher: _____ Date: _____

Exam tips (Simulation 2 is comprehensive throughout the semester and is more difficult than Simulation 1):

- Part A MC: Question 6 Note that $\text{m}^2 \rightarrow \text{cm}^2$ needs to be multiplied by 10,000; Question 11 Note that the standard amount of "remaining $\frac{1}{4}$ " is "remaining" not "all"
- Part B Calculation: Question 3: mixed numbers must be converted into improper fractions first; Question 5: remove brackets and pay attention to the distribution of negative signs
- Application of Part C: Question 1 Graph + Equation cross-topic; Question 2 Fraction + decimal unit L; Question 3 Trapezoid area + estimation + division + unit conversion four-in-one
- Make good use of 75 minutes: 15 minutes for Part A $\rightarrow$ 25 minutes for Part B $\rightarrow$ 30 minutes for Part C $\rightarrow$ 5 minutes for inspection